



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Trends That Benefit Software Developers.



Learning Outcome:

Describe common activities at work to a using short sentences in familiar spoken and written situations.

Make the result report from the analysis of the information system, based on the customer requirements, following established rules and protocols.

Can Do Statements:

- I can give details about encapsulation and trends in OOP.
- I can explain briefly the concept of encapsulation and trends.
- I can explain to a group of co-workers what encapsulation and trends are.

Ask your partner:

- What features do you think can benefit companies developing a software effective for employees?
- What could companies do to improve employees and customers' knowledge of their products and processes?





Let's Explore!

How to understand the encapsulation and its tendencies in OOP.



Instructions: Read the text below.

Hello everyone, I called this meeting because we need to focus on programming languages. There are different trends and details about encapsulation that we can use taking into account encapsulation as one fundamental OOP concept.

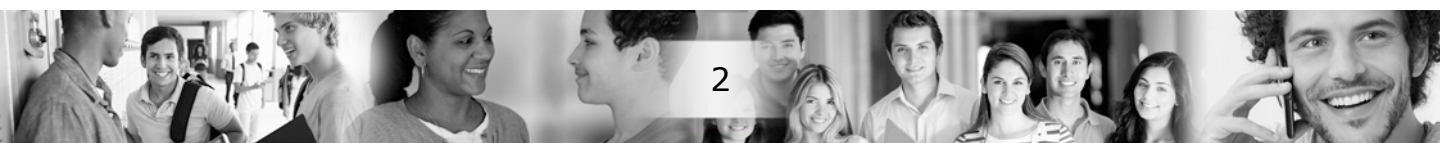
By using encapsulation, designers are able to promote the development of programs with reliability, software developers can create organized libraries and other programmers can use them. We can use interface specifications, in this way we could create more readable programs.

There are different advantages of using encapsulation and there are some large libraries created by the programmers. For example we can use this technique to hide data members and members' functions, with this, we can ensure security in our software.

We know that OOP trends have been faded by other programmers judging its capabilities and effectiveness by the implementation of Smalltalk. However, the introduction of C++ could allow programmers to improve the object-oriented world. Many software developers realized that there are many other options by using Microsoft to structure their access to a newer operating system.

With time programmers are becoming more environmentally conscious and sustainable, and they established as a trend the possibility to distribute documents electronically, which also led to office automation. Many corporate manuals are kept in the websites so people can update and consult them when necessary.

In the last years, the use of Internet has evolved and improved communication, therefore, as a company and co-workers we need to implement encapsulation processes in our day-to-day duties. In our case as a trend we can use the benefits of the Internet for internal communication and collaboration in the office, simple processes such as a blog or wikis that can benefit employees and customers' knowledge and keep everyone updated. As follows we can see the impact on customer service sales and revenue.





Let's Practice!

Details about encapsulation and trends in OOP.

Instructions: Underline all the sentences about encapsulation and trends.

1. Thanks to this processes' restrictions users are not able to manipulate objects.
2. Designers are able to promote the development of programs with reliability.
3. C++, one of the most popular languages that you can use to define objects.
4. Software developers are able to create organized libraries and other programmers can use them.
5. The ability to hide data and behavior that are not necessary to the user.
6. We can use interface specifications, in this way, we could create more readable programs.

Things people can achieve by using encapsulation	Significant elements of the encapsulation process



Let's Practice!

Explaining what encapsulation and trends are in OOP.

Instructions: Select the correct answer.

1. In this lesson Trends could refer to some modifications in
 - a. OOP
 - b. C++
 - c. Programs
 - d. HTML

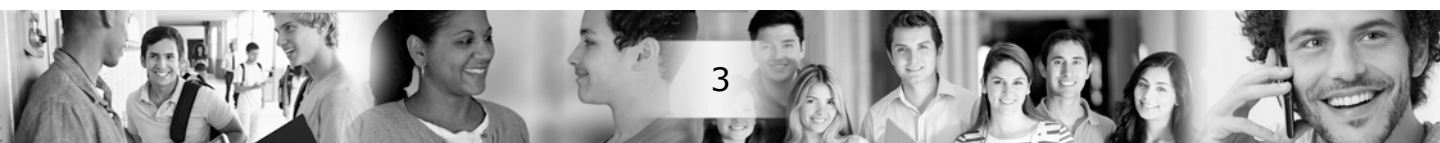
(Hint: What was the modeling language they talked about in the R2?)
2. We can use encapsulation to hide:
 - a. data
 - b. data members and members' functions
 - c. libraries
 - d. capabilities

(Hint: Important information about people involved in our companies.)
3. Thanks to encapsulation you can get protection of the program data from accidental
 - a. process
 - b. Modifications
 - c. transference
 - d. users

(Hint: synonym of transformation)
4. Encapsulation is able to add or modify internal structure through a request by encapsulating.
 - a. OOP
 - b. class
 - c. C++
 - d. internal communication

(Hint: The smaller parts of OOP)
5. One of the principal advantages of encapsulation could be that:
 - a. it improves the access to cookies
 - b. It decreases the time to develop a software.
 - c. It organizes libraries and other programmers to use them.
 - d. It isolates an object in a way that you can't change it.

(Hint: It helps you avoid modifying an object by mistake.)





Extension Activity

Talking about encapsulation in OOP.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about encapsulation in OOP.
- Describe how it works.
- Make a brochure about encapsulation.
- Show your brochure to your classmates and instructor.

Self-Assessment

- | | Yes | No |
|--|-----------------------|-----------------------|
| 1. I look for specific information related to OOP. | ☐ | ☐ |
| 2. I describe what I understood. | ☐ | ☐ |
| 3. I make a brochure. | ☐ | ☐ |
| 4. I show my brochure. | ☐ | ☐ |



Target Words

- **wiki** (noun): a website that allows users to add, delete, and edit the content, or the information of a website
e.g. We can use the benefits of the Internet for internal communication and collaboration in the office simple processes such as a blog or wikis that can benefit employees and customers' knowledge and keep everyone updated.
- **Environmentally conscious** (noun): been worried about the eco system to keep it safe
e.g. With time programmers are becoming more environmentally conscious and sustainable.
- **sustainable** (adjective): able to continue over a period of time, causing little or no damage to the environment
e.g. With time programmers are becoming more environmentally conscious and sustainable.
- **promote** (verb): to encourage people to, use, do, or support something
e.g. By using encapsulation, designers are able to promote the development of programs with reliability.
- **reliability** (noun): applied to objects, a person or an idea capable of being relied on
e.g. By using encapsulation, designers are able to promote the development of programs with reliability.
- **fad** (noun): critics to a style, activity, or interest that is very popular for a short period of time
e.g. We know that OOP trends have been fad by other programmers judging its capabilities and effectiveness by the implementation of Smalltalk.



Learning Points

Modal can for ability and possibility.

Modal verbs are auxiliary verbs, which need another verb to function. There are three verbs that you can use to express possibility or ability.

The modal verbs to talk about possibility or ability are **can**, **be able to**, and **could**.

You can use these modal verbs in the affirmative form like this:

Subject + **modal verb** + **main verb** + complement.

Be able to needs to modify the verb **to be** according to the subject. (am not, isn't, aren't)

Remember the main uses for these modal verbs:
(Chart 1)

Example:

*Software developers **can** work with different programs according to their needs.*

*We **are able to** understand how to use UML and other languages.*

*They **could** use a different type of diagram.*

CAN	Express ability or show ability To suggest a possibility or give an option To ask for or to give permission (informal) To show impossibility (can't)
BE ABLE TO	We use it to talk about ability, possibility and capability.
COULD	To show past ability (past of the verb can) To ask a polite question To use a more formal way than with can To suggest a possibility/opportunity or give an option.

Examples

- *There are different trends and details about encapsulation that we can use taking into account encapsulation as one fundamental OOP concept.*
- *By using encapsulation, designers are able to promote the development of programs with reliability*
- *In this way we could create more readable programs.*



Learning Points

Singular and plural of countable nouns.

Countable nouns are the ones that you can use numbers to count instead of a different measurement unit; they are individual things, they can be singular or plural.

You can use **countable nouns** with *a/an*, and with numbers and other determiners (quantifiers, demonstratives, articles, possessive adjectives).

(Chart 1)

For the plural form of most nouns, add **s**.

pencil – pencils
desk – desks
program – programs

For nouns that end in ch, x, s, or s sounds, add **es**.

box – boxes
watch – watches
bus – buses

For nouns ending in f or fe, change f to v and add **es**.

wolf – wolves
leaf – leaves
life – lives

Quatifiers	Demostratives	Articles	Possessive adjectives
few, some, many	this, these, those, that	the, a, an	my, your , his, her, our, their, its.